

Sourabh Gome

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Summary

Senior Software Engineer with 4.5+ years of experience designing scalable distributed systems and device infrastructure at BrowserStack. Proven track record of improving performance (30%+), scaling systems (5×), and delivering reliable high-throughput solutions across large device fleets. Experienced in system design, backend architecture, and building AI-powered developer tooling using RAG pipelines.

Experience

BrowserStack

Apr 2025 – Present

Senior Software Engineer

- Designed and built an internal **RAG-based debugging assistant** integrated with Slack, leveraging vector embeddings over internal knowledge bases; enabled automated diagnostics using **Model Context Protocol (MCP)** integrations, reduced debugging time for engineers, and lowered operational load by **50%** by empowering customer engineers to independently resolve issues
- Architected and redesigned VoiceOver automation system by removing host version dependency, improving scalability by **5×**; incorporated reliability and fault-tolerant communication workflows to ensure consistent execution across distributed device infrastructure
- Reduced Android device cleanup latency from **215s to 172s (20%)** and improved median by **30%** by optimizing backend workflows and parallelizing operations

BrowserStack

Oct 2021 – Mar 2025

Software Engineer

- Designed and implemented a scalable Apple Pay validation framework with fallback mechanisms, improving card success accuracy by **95%**
- Built a distributed TestFlight installation system eliminating App Store dependency, enabling execution across **thousands of devices** with improved fault tolerance
- Developed Android video injection system using low-level API interception, enabling real-time media manipulation across device environments
- Designed browser-level video injection architecture after evaluating trade-offs between performance, reliability, and maintainability
- Enabled parallel execution of device patching features, improving concurrency and reducing test setup time
- Built Apple Pay automation using OpenCV-based UI detection integrated with WebDriverAgent, improving reliability of automation flows
- Delivered Safari settings control system with backend coordination, improving bandwidth efficiency and session performance
- Debugged distributed system failures during on-call rotations and improved observability across multiple services

Artha Energy Resources

Sep 2020 – Jun 2021

Software Engineer Intern

- Developed backend components and data models to improve database interaction and system responsiveness
- Built frontend integrations to visualize API responses, improving usability and debugging workflows
- Collaborated using Git-based workflows for efficient code integration

Technical Skills

- **Languages:** Java, Python, C++, JavaScript, Swift, Objective-C
- **System Design:** Distributed Systems, Scalability, High Availability, Fault Tolerance, Concurrency, API Design
- **Backend:** Spring Boot, REST APIs, Microservices Architecture
- **Databases:** MySQL, PostgreSQL, MongoDB
- **AI/ML:** Retrieval-Augmented Generation (RAG), Vector Databases, Embeddings, LLM-based Systems, Model Context Protocol (MCP)
- **Tools:** Git, OpenCV, Xcode, Android Studio

Projects

- **Path Finding Visualizer**
Built an interactive web application to visualize graph algorithms such as Dijkstra, BFS, and DFS
- **Remote Desktop Manager**
Developed a multi-threaded application using socket programming to enable remote device control and file transfer

Education

Ujjain Engineering College
B.Tech in Computer Science

2017 – 2021

Achievements

- Received **Champions Award Twice** for outstanding engineering performance at BrowserStack
- Recognized with **Blackbelt Nomination** for technical excellence
- Solved **1000+ problems** on LeetCode and achieved **3-star rating (1726)** on CodeChef
- Cleared Google Code Jam (2019, 2020) qualification rounds